A Positively Curved Seven-Manifold Without Effective Smooth Circle Actions

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Abstract

We construct a compact Riemannian manifold M with strictly positive sectional curvature and prove that M admits no effective smooth S^1-action, which yields to a counterexample of a longstanding conjecture attributed to Yau.

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1 Yau's circle-action conjecture

A long-standing question in positive sectional curvature asks whether positive curvature forces the existence of continuous symmetry. The formulation used in this paper is the following.

Conjecture 1.1 (Yau's circle-action conjecture)**.** *Every closed smooth manifold that admits a Riemannian metric with strictly positive sectional curvature admits a nontrivial effective smooth action of the circle group S^1.*

The question is often stated by saying that every closed positively curved manifold should admit "some smooth circle group action." It is important to distinguish this smooth formulation from the stronger requirement that the given positively curved metric itself be invariant under the circle. A smooth action need not preserve a prescribed metric: averaging an arbitrary auxiliary metric over the circle produces an invariant metric, but it need not retain positive sectional curvature. The conjecture above asks only for a smooth action on the underlying manifold.

The problem is included in Yau's collection of open problems in geometry [8]; Yau later summarized the same motivation by saying that each positively curved manifold should admit

some smooth circle group action [7]. The purpose of this paper is to give a completely explicit counterexample to the unrestricted formulation above.

Theorem 1.2. *There exists a closed, connected, smooth seven-manifold M with the following properties:*

(i) $\pi_1(M) \cong (\mathbb{Z}/6)^2$*;*

(ii) M admits a Riemannian metric with strictly positive sectional curvature;

(iii) M admits no effective smooth action of S^1.

Consequently, M is a counterexample of conjecture 1.1.

For clarity, the conclusion concerns arbitrary smooth circle actions, not merely isometric actions for the displayed positive-curvature metric. The manifold is not simply connected. Thus theorem 1.2 disproves the no-hypothesis version of the conjecture, while leaving open any strengthened version in which simple connectivity is imposed from the outset.

2 The explicit positively curved quotient

Put

$$P = SU(3), \qquad H = \{\mathrm{diag}(z, z, z^{-2}) : z \in S^1\}, \qquad K = S(U(2) \times U(1)).$$

Then H is central in K, and $K/H \cong PU(2) \cong SO(3)$ acts freely on the right of $X = P/H$.

Let $\omega = e^{2\pi i/3}$ and define

$$u = \mathrm{diag}(1, \omega, \omega^2), \qquad ve_1 = e_2, \quad ve_2 = e_3, \quad ve_3 = e_1.$$

The images of u and v in $PSU(3)$ generate a group $A \cong (\mathbb{Z}/3)^2$. Inside $K/H \cong SO(3)$, let $B \cong (\mathbb{Z}/2)^2$ be generated by the half-turns about two orthogonal axes. The left A-action and the right B-action commute. Define

$$\Gamma = A \times B, \qquad M = X/\Gamma.$$

Proposition 2.1. *The action of Γ on X is free. The manifold M is closed, connected, smooth, and seven-dimensional, has fundamental group $(\mathbb{Z}/6)^2$, and admits a metric of strictly positive sectional curvature.*

Proof. The space X is the Aloff–Wallach space $N_{1,1}$. It has dimension $8 - 1 = 7$. The last-column map gives a fibration $S^3 \longrightarrow SU(3) \longrightarrow S^5$, so $SU(3)$ is simply connected. Since H is connected, the homotopy sequence of $H \to P \to X$ gives $\pi_1(X) = 0$.

The left $SU(3)$-action on X has kernel $Z(SU(3)) \cong \mathbb{Z}/3$, and the projective Heisenberg group generated by u and v has projective image A. Every noncentral element of the group generated by u, v has three distinct eigenvalues (a scalar multiple of $1, \omega, \omega^2$), whereas every element of H has a repeated eigenvalue. Therefore no nonidentity element of A is conjugate into H, and the left A-action on X is free.

The right action of B is free because it is the restriction of the free right action of K/H. If $(a, b) \in A \times B$ fixes a point, then its third power is $(1, b)$, which fixes the same point. Freeness of the right B-action gives $b = 1$, and then freeness of the left A-action gives $a = 1$. Thus Γ acts freely.

The Aloff–Wallach space X admits a strictly positively curved metric invariant under left $SU(3)$ and right K/H. Concretely, one may use the Eschenburg metric with weights $(1, 1, 1)$ and $(0, 0, 3)$: its defining circle action on $SU(3)$ is

$$q \longmapsto (zI)q\,\mathrm{diag}(1, 1, z^{-3}) = q\,\mathrm{diag}(z, z, z^{-2}),$$

so the quotient is exactly X; the metric upstairs is left $SU(3)$-invariant and right K-invariant [1, 4]. It is therefore invariant under Γ, and the quotient metric on M has strictly positive sectional curvature because the quotient map is a local isometry.

Finally, X is simply connected and Γ acts freely, so $X \to M$ is the universal covering and

$$\pi_1(M) \cong \Gamma \cong (\mathbb{Z}/3)^2 \times (\mathbb{Z}/2)^2 \cong (\mathbb{Z}/6)^2.$$

□

3 Proof strategy: why M is a counterexample

The proof of the main theorem is organized around a contradiction. Assume that M admits an effective smooth S^1-action. Because X is simply connected and $\Gamma = A \times B$ acts freely, the action lifts, after a finite covering of the parameter circle, to an effective circle action β on X commuting with both A and B. The principal bundle $P = SU(3) \to X$ then allows us to lift β to a periodic action and to place it in a two-torus together with the principal circle H.

The rest of the argument has two independent halves, linked by the same circle action β:

1. Sections 4 and 5 establish the lifting statements and the fixed-set restrictions on $P = SU(3)$. The key input is $H^*(SU(3); \mathbb{Z}) = \Lambda(a_3, b_5)$ together with the nonzero relation $Sq^2 a_3 = b_5$.

2. The left action of $A \cong (\mathbb{Z}/3)^2$ is used in Section 6. It first rules out fixed points of β, so β is locally free; it then implies that the unique involution of β acts freely.

3. The right action of $B \cong (\mathbb{Z}/2)^2$ is used in Section 7. A Borel spectral-sequence calculation and the extension $1 \to H \to J \to B \to 1$ show that the involution of every circle action commuting with B has a fixed point.

For the lifted action commuting with $A \times B$, the involution in the last two statements is the same involution. It would therefore have to be both free and nonfree, which is impossible. Section 8 records the formal completion of this contradiction.

4 Lifting circle actions

We first record the two lifting statements used throughout the proof.

Lemma 4.1. *Let a finite group D act freely on a connected, simply connected, closed manifold Y. Every effective smooth circle action on Y/D lifts, after a finite covering of the parameter circle, to an effective smooth circle action on Y commuting with D. The fixed set upstairs is the inverse image of the fixed set downstairs.*

Proof. Lift the generating vector field of the action on Y/D through the covering map $Y \to Y/D$. The lifted vector field is D-invariant and complete, so its flow Φ_t commutes with D. The time-one map is a deck transformation z; because the deck group is finite, z has finite order, say m, and $\Phi_m = \mathrm{id}$. Thus the lifted flow defines a circle action after replacing the parameter circle $\mathbb{R}/\mathbb{Z}$ by $\mathbb{R}/m\mathbb{Z}$.

If $\Phi_t = \mathrm{id}$, effectiveness downstairs implies that t is an integer in the original parameter, and the relation with the deck transformation shows that the resulting lifted action is effective after the finite reparametrization. A point above a fixed point downstairs has orbit contained in a discrete covering fiber, hence is fixed upstairs; the converse follows by projection. □

Let $\pi : P \to X$ denote the principal H-bundle. The next lemma makes the lift periodic on P and allows us to combine it with the principal circle.

Lemma 4.2. *Suppose a finite group D acts on X and that there is a compact central extension*

$$1 \longrightarrow H \longrightarrow E \longrightarrow D \longrightarrow 1$$

acting on P by principal bundle automorphisms, with H acting as the principal circle. If an effective circle β on X commutes with D, then β admits a period-one lift Ψ to P commuting with E. The circles H and Ψ generate an effective two-torus. If F is a component of X^β, the lift may be chosen to fix $\pi^{-1}(F)$ pointwise.

Proof. Choose an E-invariant connection on P by averaging any connection. Because E is a central extension, its conjugation action on the principal circle H is trivial; since β commutes with D, the horizontal lift of its generating vector field therefore commutes with E. Let Φ_t denote this horizontal flow. Since Φ_1 projects to the identity on X, it is a gauge transformation

$$\Phi_1(p) = p\, q(\pi(p)), \qquad q : X \to H.$$

Because X is simply connected, write $q(x) = h(f(x))$ for a smooth real-valued function f and a fixed parametrization $h : \mathbb{R}/\mathbb{Z} \to H$. Commutation of Φ_1 with the flow implies that q is β-invariant, hence $f(\beta_t x) - f(x)$ is an integer-valued continuous function of t which vanishes at $t = 0$. It is therefore zero. Similarly, for $d \in D$, the difference $f(dx) - f(x)$ is an integer constant on connected X; summing this constant over the finite order of d gives zero.

Define

$$\Psi_t(p) = \Phi_t(p) h\big(-t f(\pi(p))\big).$$

The invariance of f gives the group law, and $\Psi_1 = \mathrm{id}$. The flow commutes with E. It is effective, and the joint action of H and Ψ is effective because a relation $R_s \Psi_t = \mathrm{id}$ projects to $\beta_t = \mathrm{id}$ and then reduces to freeness of the principal H-action.

Over a fixed component F, the horizontal flow is stationary, so $q|_F = 1$ and $f|_F$ is an integer constant. Multiplying Ψ_t by the corresponding principal rotation makes the lift fix $\pi^{-1}(F)$ pointwise. □

For the left action we use the central extension

$$E_A = (\widetilde{A} \times H)/\{(c, c) : c \in Z_3\},$$

where $\widetilde{A} = \langle u, v \rangle$ is the order-27 Heisenberg group. The class $[(g, h)]$ acts on P by $p \mapsto gph^{-1}$, and the map $E_A \to A$ sends $[(g, h)]$ to the projective class of g; its kernel is the principal circle H. For the right action, let

$$J = H \cdot Q_8 \subset K, \qquad 1 \longrightarrow H \longrightarrow J \longrightarrow B \longrightarrow 1.$$

5 The cohomological fixed-set toolkit on $SU(3)$

5.1 The cohomology ring and the Steenrod square

The fibration $S^3 \to SU(3) \to S^5$ gives

$$H^*(P; \mathbb{Z}) = \Lambda(a_3, b_5), \qquad |a_3| = 3, \quad |b_5| = 5,$$

and, with coefficients in $\mathbb{F}_2$,

$$Sq^2 a_3 = b_5 \neq 0. \tag{1}$$

One way to see the operation is to use the universal fibration $P \to EP \to BP$. The classes a_3 and b_5 transgress to c_2 and c_3, and the splitting principle gives

$$Sq^2 c_2 = c_1 c_2 + c_3 = c_3$$

for $SU(3)$, because $c_1 = 0$. The Steenrod transgression theorem then yields (1). In particular,

$$\sum_j \dim_{\mathbb{F}_p} H^j(P;\mathbb{F}_p) = 4$$

for every prime p.

5.2 Some facts in Smith's theory

We use the standard coefficient rings

$$H^*(BS^1;\mathbb{F}_2) = \mathbb{F}_2[s], \quad |s| = 1, \qquad H^*(B(S^1)^2;\mathbb{F}_2) = \mathbb{F}_2[u,v], \quad |u| = |v| = 2,$$

$$H^*(B(\mathbb{Z}/2)^2;\mathbb{F}_2) = \mathbb{F}_2[s_1,s_2], \quad |s_i| = 1, \qquad H^*(B(\mathbb{Z}/3)^2;\mathbb{F}_3) = \mathbb{F}_3[t_1,t_2] \otimes \Lambda(s_1,s_2).$$

For a compact group G acting on Y, write $Y_G = EG \times_G Y$. The Borel fibration $Y \to Y_G \to BG$ has the multiplicative cohomological Serre spectral sequence.

We will also use the following standard consequences of Smith theory and localization.

Lemma 5.1. *Let $D \neq \varnothing$ be the fixed set in P of either a nontrivial circle action or an involution contained in a smooth torus action, and assume that the action commutes with the free principal H-action. Then D is connected and has dimension 2, 4, or 6. In the circle case it is orientable and has total Betti number 4 over every prime field. In the involution case its total mod-two Betti number is 4.*

Proof. Choose a metric invariant under the acting compact group and H. Fixed components of a circle have even codimension and oriented normal bundles, so they are orientable. For an involution inside a torus, the torus action preserves orientation, and the normal representation is the (-1)-eigenspace, whose real dimension is therefore even.

The free H-action preserves every fixed component. Thus every component has Euler characteristic zero and positive even dimension. Over $\mathbb{F}_2$, and over any prime field in the orientable case, duality gives at least two even-degree and at least two odd-degree classes. Each component therefore contributes at least four to the total Betti number. For a circle action, choose a prime p larger than the orders of all finite stabilizers (these orders are bounded for a smooth action on a compact manifold). Then the fixed set of a subgroup $C_p \subset S^1$ is exactly the circle fixed set, so Smith theory gives the same bound; for an involution use $p = 2$. Since $b(P;\mathbb{F}_p) = 4$, the fixed set has exactly one component and total Betti number four. A fixed component of dimension eight would contain an open set, forcing the action to be trivial. Thus only dimensions 2, 4, and 6 occur. □

Lemma 5.2. *The fixed set of an orientation-preserving involution on $P = SU(3)$ is orientable.*

Proof. Since P is a simply connected Lie group, it is parallelizable and has a unique spin structure. Every element of the acting torus is isotopic to the identity, so the involution preserves both the orientation and this spin structure. Edmonds's orientability theorem for orientation- and spin-structure-preserving involutions now implies that every component of the fixed set is orientable [2]. □

Lemma 5.3. *The fixed set in lemma 5.1 cannot have dimension six.*

Proof. First consider an involution σ. Put $R = \mathbb{F}_2[s]$, $|s| = 1$. Since the involution is isotopic to the identity in the torus, the Borel spectral sequence starts with $E_2 = R \otimes \Lambda(a_3, b_5)$. Localization and the equality in Smith's bound identify the R-rank of the abutment with $b(D;\mathbb{F}_2) = 4$. If a first nonzero differential existed, all earlier pages would still be free of rank four over R; tensoring with the fraction field $\mathbb{F}_2(s)$ would preserve that differential and strictly

lower the total rank, contradicting localization. Therefore the spectral sequence collapses. Its fiber edge map is onto, so Leray–Hirsch makes the equivariant cohomology a free R-module. Fixed-set restriction is injective because its kernel is s-torsion.

Suppose $D = P^\sigma$ had dimension six. Its total Betti number is four and $\chi(D) = 0$. Thus $b_2(D) = b_4(D) = 0$, and duality gives either $(b_1, b_3) = (0, 2)$ or $(1, 0)$. Let α be a lift of a_3 whose restriction has no base-only term. In the first case, Sq^2 of the restriction vanishes for dimensional reasons, contradicting injectivity and $Sq^2 a_3 = b_5$. In the second case, if z generates $H^1(D; \mathbb{F}_2)$, then the restriction is $s^2 z$, and

$$Sq^2(s^2 z) = s^4 z.$$

This forces $Sq^2\alpha = s^2\alpha$, which again contradicts the restriction to the fiber because s restricts to zero while $Sq^2 a_3 = b_5 \neq 0$.

Now suppose D is the fixed set of an effective circle and has dimension six. The normal representation is a real two-plane of nonzero weight w. If $|w| > 1$, a nonidentity circle element would fix a point and have identity derivative, hence would be the identity after averaging a metric, a contradiction. Thus $w = \pm 1$. The involution acts by $-I$ on the normal plane, so D is a component of the fixed set of that involution. Smith equality and the free H-action exclude additional components, reducing to the involution case just ruled out. □

6 The left factor forces the lifted involution to be free

We first isolate the three-dimensional obstruction used to rule out fixed points of a circle commuting with A.

Lemma 6.1. *No connected closed oriented three-manifold whose mod-three cohomology ring is that of S^3 or $S^1 \times S^2$ admits a free action of $A = (\mathbb{Z}/3)^2$.*

Proof. In either case the action of A on the cohomology is trivial: any homomorphism from the 3-group A to $\mathbb{F}_3^\times \cong \mathbb{Z}/2$ is trivial. If the action were free, the Borel construction would be homotopy equivalent to the quotient, whose cohomology vanishes above degree three.

For a mod-three cohomology sphere, the only possible differential is $d_4(w) = c$ for the degree-three fiber generator. The row-zero quotient contains $\mathbb{F}_3[t_1, t_2]/(\bar{c})$ after killing the exterior generators. This is infinite-dimensional for every polynomial $\bar{c}$, since one nonzero polynomial in two variables cannot contain powers of both t_1 and t_2 in the ideal it generates. This contradicts finite-dimensionality of the quotient.

For $S^1 \times S^2$, write

$$H^*(Y; \mathbb{F}_3) = \Lambda(x) \otimes \mathbb{F}_3[y]/(y^2), \qquad |x| = 1, \quad |y| = 2.$$

Write $d_2(x) = a$ and $d_2(y) = bx$. The relation $y^2 = 0$ gives $2bxy = 0$, hence $b = 0$. The E_3 rows are then described by $R/(a)$ and $\mathrm{Ann}_R(a)$, where $R = H^*(BA; \mathbb{F}_3)$. Multiplication by y pairs the rows. The derivation rule and the relation $y^2 = 0$ force all possible later differentials to vanish. Thus row zero still contains $R/(a)$, whose quotient by the exterior ideal is $\mathbb{F}_3[t_1, t_2]/(\bar{a})$ and is infinite-dimensional. Again this contradicts the finite-dimensional cohomology of the quotient. □

Proposition 6.2. *Every effective smooth circle action on X commuting with A is locally free.*

Proof. Suppose that a component F of the circle fixed set is nonempty. Apply lemma 4.2 to the extension E_A and choose a lifted circle T_0 fixing $\pi^{-1}(F)$ pointwise. Its fixed set $D = P^{T_0}$ is connected, orientable, has total Betti number four, and by lemma 5.3 has dimension two or four. Since $\pi(D)$ is contained in the fixed set downstairs and contains F, we have $D = \pi^{-1}(F)$.

The A-action preserves F and is free. The orientation of D together with the oriented circle fibers gives an orientation on F.

If $\dim D = 2$, then F is a circle. A finite free group of orientation-preserving diffeomorphisms of a circle is cyclic, so it cannot contain $A \cong (\mathbb{Z}/3)^2$.

Suppose $\dim D = 4$. We record the small Gysin calculation used here. Let $b = b_1(F;\mathbb{F}_3)$ and let $e \in H^2(F;\mathbb{F}_3)$ be the Euler class of the principal circle bundle $D \to F$. If $e = 0$, the Gysin sequence gives $b_1(D) = b + 1$ and $b_2(D) = 2b$. If $e \neq 0$, Poincare duality on the oriented three-manifold F gives $b_1(D) = b$ and $b_2(D) = 2b - 2$. Since lemma 5.1 gives $b(D) = 4$ and $\chi(D) = 0$, the only possibilities are $(e = 0, b = 0)$ or $(e \neq 0, b = 1)$; in either case

$$b_1(D) = 1, \qquad b_2(D) = 0.$$

The Gysin sequence now gives $b_2(F;\mathbb{F}_3) \leq 1$. By three-dimensional duality, either F is a mod-three cohomology sphere or its cohomology ring is that of $S^1 \times S^2$. Both cases are excluded by lemma 6.1. Therefore the circle action has no fixed point and is locally free. □

Proposition 6.3. *Let β be an effective, locally free, smooth circle action on X commuting with A. Its unique involution acts freely.*

Proof. Use lemma 4.2 to obtain a period-one lift Ψ commuting with E_A. Suppose the involution of β fixes $x \in X$. On the principal fiber over x, the half-period map $\Psi_{1/2}$ is a principal rotation whose square is the identity. Hence one of

$$\sigma = \Psi_{1/2}, \qquad \sigma = R_{1/2}\Psi_{1/2}$$

fixes the entire principal fiber. Both are nonidentity involutions in the torus generated by H and Ψ.

Let $D = P^\sigma$. By lemmas 5.1 to 5.3, D is connected, orientable, and has dimension two or four, with total mod-two Betti number four. It is preserved by E_A, H, and Ψ. Set $Y = D/H$. Then Y is a connected closed oriented manifold with a free A-action and a locally free circle action; the freeness follows because Y embeds in X and the A-action on X is free.

The case $\dim D = 2$ would make Y a circle, which is impossible for a free A-action. Suppose $\dim D = 4$. The same Gysin calculation as in the proof of proposition 6.2, now with coefficients in $\mathbb{F}_2$, shows

$$b_1(D;\mathbb{F}_2) = 1, \qquad H^2(D;\mathbb{F}_2) = 0.$$

Consequently the Gysin sequence and duality imply

$$b_1(Y;\mathbb{Q}) \leq b_1(Y;\mathbb{F}_2) = b_2(Y;\mathbb{F}_2) \leq 1.$$

After dividing the restricted circle by its finite global kernel, write the effective circle as T_0. Its orbit space $S = Y/T_0$ is a connected closed oriented surface. Since the stabilizers are finite, the Borel comparison over $\mathbb{Q}$ gives $H^*(S;\mathbb{Q}) \cong H^*(Y_{T_0};\mathbb{Q})$. In total degree one, the Borel spectral sequence yields

$$b_1(S;\mathbb{Q}) \leq b_1(Y;\mathbb{Q}) \leq 1.$$

Thus S has genus zero and $\chi(S) = 2$.

On the other hand, every stabilizer of the induced A-action on S is cyclic. If a fixes an orbit $[y]$, then $ay = ty$ for some $t \in T_0$; the resulting map from the stabilizer into $T_0/T_{0,y}$ is injective because A acts freely on Y, so the stabilizer acts by translations on the orbit circle. Since A has exponent three, each stabilizer has order one or three. An equivariant finite CW decomposition of the compact smooth $A \times T_0$-manifold Y (and hence of its orbit surface) therefore gives cells whose T_0-quotients occur in A-orbits of size nine or three [5]. Consequently $3 \mid \chi(S)$, contradicting $\chi(S) = 2$. Hence the involution is free. □

7 The right factor forces a fixed point

All cohomology in this section has coefficients in $\mathbb{F}_2$.

Proposition 7.1. *Every effective smooth circle action on X commuting with B has an involution fixed point.*

Proof. Assume, for contradiction, that the involution of β is free. Then every stabilizer of β is finite of odd order: any finite stabilizer of even order would contain the unique involution. Apply lemma 4.2 to the extension J and obtain a lifted circle T' commuting with J. Let $T = H \times T'$ act on P. The projection of a T-stabilizer into T' is injective because a principal rotation fixing a point is the identity. Hence all T-stabilizers have odd order. Borel comparison gives

$$H^*(P_T;\mathbb{F}_2) \cong H^*(P/T;\mathbb{F}_2), \qquad H^j(P_T;\mathbb{F}_2) = 0 \quad (j > 6). \tag{2}$$

Let $u, v \in H^2(BT;\mathbb{F}_2)$ correspond to H and T'. With $R = \mathbb{F}_2[u,v]$, the Borel spectral sequence begins with $E_2 = R \otimes \Lambda(a_3, b_5)$. The first possible differential on a_3 is d_4, while the first possible differential on b_5 is d_6; a potential d_3 from degree five to degree three has target in odd base degree and vanishes. Restricting to H means setting $v = 0$. The weights $(1, 1, -2)$ give $c_2|_{BH} = (1 - 2 - 2)u^2 = u^2$ over $\mathbb{F}_2$. Therefore

$$d_4(a_3) = q = u^2 + \epsilon uv + \eta v^2, \qquad \epsilon, \eta \in \mathbb{F}_2. \tag{3}$$

By (1) and the Steenrod transgression theorem,

$$d_6(b_5) = Sq^2 q = \epsilon(u^2 v + uv^2) \pmod{(q)}. \tag{4}$$

The polynomial q is nonzero, so multiplication by q is injective on R. After d_4, only rows zero and five remain, both with coefficient module $R/(q)$. If $\epsilon = 0$, then $d_6 = 0$. If $(\epsilon, \eta) = (1, 0)$, then $q = u^2 + uv$ and $u^2 v + uv^2 = vq$, so again $d_6 = 0$ modulo (q). In either case row zero remains infinite-dimensional, contradicting (2). Hence

$$q = u^2 + uv + v^2.$$

This quadratic is irreducible over $\mathbb{F}_2$, so $R/(q)$ is a domain. The element $r = u^2 v + uv^2$ is nonzero modulo (q) and hence is not a zero-divisor. It kills the row-five class, so only the filtration-zero row survives. The edge map from $H^*(BT)$ supplies all surviving classes, and there is no higher-filtration piece in the same total degree that could produce an additive or multiplicative extension. Consequently

$$H^*(P_T;\mathbb{F}_2) \cong \mathbb{F}_2[u,v]/(u^2 + uv + v^2,\ u^2 v + uv^2). \tag{5}$$

The surviving classes are represented by $1, u, v, uv, v^2, uv^2$. Now set $L = J \times T'$ and note that $L/T \cong B$. The action gives a fibration

$$P_T \longrightarrow P_L \longrightarrow BB.$$

Because the lifts of elements of B commute with T, the local coefficient system on the fiber cohomology is trivial. Write $H^*(BB;\mathbb{F}_2) = \mathbb{F}_2[s_1, s_2]$, $|s_i| = 1$. The extension $1 \to H \to J \to B \to 1$ is the pullback of $S^1 \to U(2) \to SO(3)$. The universal calculation gives $d_3(u) = w_3$, and the three real line characters of the B-representation are s_1, s_2, and $s_1 + s_2$. Hence

$$d_3(u) = k = s_1 s_2 (s_1 + s_2), \qquad d_3(v) = 0. \tag{6}$$

Naturality transfers (6) to the upper fibration. Since the fiber has no odd-degree cohomology, $d_2 = 0$, and hence

$$E_3 = E_2 = \mathbb{F}_2[s_1, s_2] \otimes H^*(P_T;\mathbb{F}_2).$$

Applying the derivation d_3 to the actual relation in (5) gives

$$0 = d_3(u^2 + uv + v^2) = ku + uk + kv = kv.$$

But $kv \neq 0$ on this tensor-product page: $k \neq 0$ in $\mathbb{F}_2[s_1, s_2]$ and $v \neq 0$ in the fiber ring. This contradiction proves that the involution of β cannot be free. □

8 Completion of the proof

Proof of theorem 1.2. The geometric assertions and the fundamental group are established in proposition 2.1. Suppose that M admitted an effective smooth circle action. By lemma 4.1, after a finite covering of the parameter circle it would lift to an effective circle action β on X commuting with $A \times B$.

Since β commutes with A, proposition 6.2 shows that β is locally free. Then proposition 6.3 says that its unique involution is free. Since β also commutes with B, proposition 7.1 says that the same involution has a fixed point. This is impossible. Therefore M admits no effective smooth circle action. □

Remark 8.1 (Scope of the counterexample)**.** *The construction refutes the unrestricted formulation of Yau's circle-action conjecture. It does not refute a variant in which the positively curved manifold is required to be simply connected, because $\pi_1(M) \cong (\mathbb{Z}/6)^2$. The obstruction is nevertheless stronger than the absence of symmetry for the particular homogeneous metric: it rules out every effective smooth circle action on the underlying manifold, and therefore rules out positive-dimensional compact Lie group actions and positive-dimensional isometry groups for all Riemannian metrics.*

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Statement on the Use of Artificial Intelligence

HyProver, an AI system developed by Tencent Hunyuan, generated the initial proofs of the main results presented in this paper. The authors subsequently reviewed and verified the arguments, revised the proofs, and rewrote the exposition. The final presentation is the responsibility of the authors.